

SIMATS ENGINEERING

Department of Computer Science & Engineering

CO5-ClassActivity

Programming in Java - File I/O Operations

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Course Code	CSA0904
Course Name	Programming in Java
Assignment Topic	CO5-ClassActivity: File Reading and Console Display
Question No.	Question 2

QUESTION 2: READ CONTENT FROM FILE AND DISPLAY ON CONSOLE

Problem Statement:

Create a Java program to read the contents of an existing file (output.txt) line-by-line using character streams (FileReader and BufferedReader) and display each line with line numbers on the console.

Source Code (ReadAndDisplayFile.java):

```
import java.io.BufferedReader;
import java.io.File;
import java.io.FileReader;
import java.io.IOException;

public class ReadAndDisplayFile {
    public static void main(String[] args) {
        String fileName = "output.txt";
        File file = new File(fileName);

        // Check if file exists before reading
        if (!file.exists()) {
            System.out.println("Error: The file '" + fileName + "' does not exist.");
            return;
        }

        System.out.println("--- Reading Content from: " + fileName + " ---");

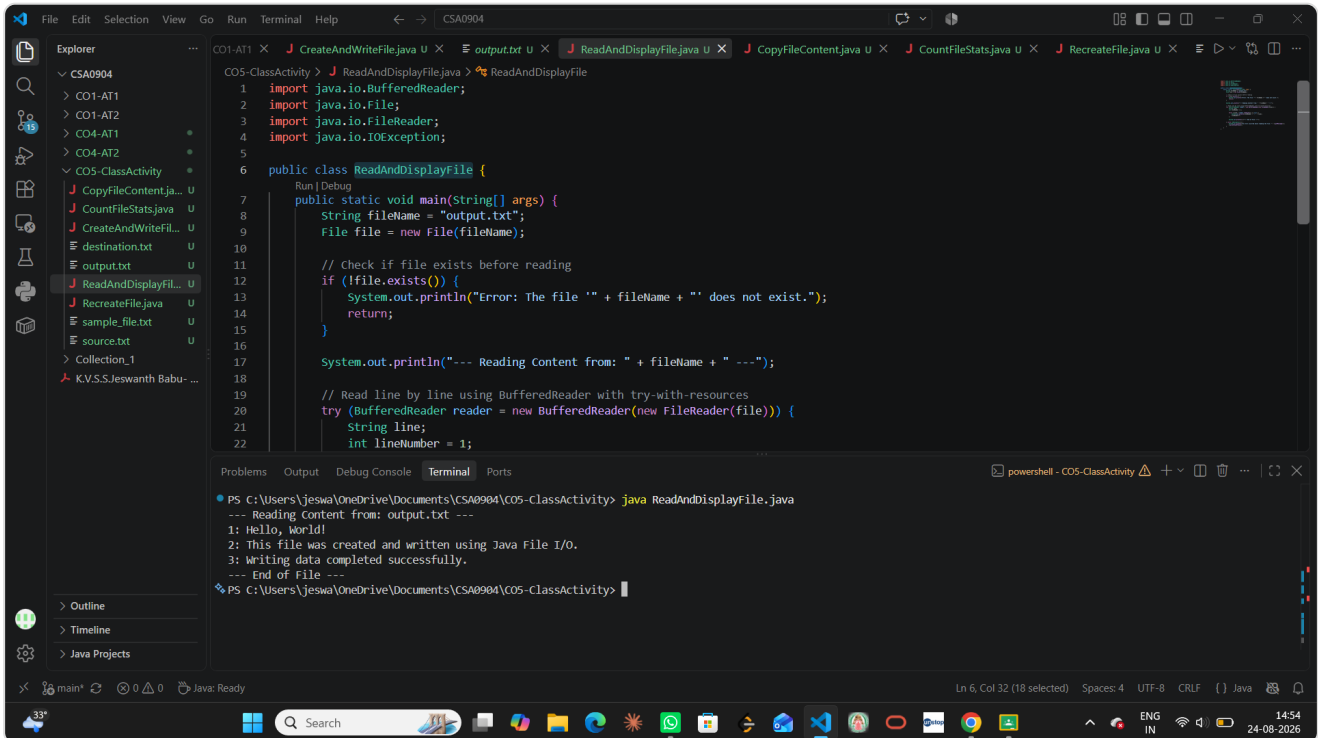
        // Read line by line using BufferedReader with try-with-resources
        try (BufferedReader reader = new BufferedReader(new FileReader(file))) {
            String line;
            int lineNumber = 1;

            while ((line = reader.readLine()) != null) {
                System.out.println(lineNumber + ": " + line);
                lineNumber++;
            }

            System.out.println("--- End of File ---");
        } catch (IOException e) {
            System.err.println("An error occurred while reading the file: " + e.getMessage());
            e.printStackTrace();
        }
    }
}
```

EXECUTION & VERIFICATION OUTPUT

Program Execution & Terminal Output:



The screenshot displays an IDE window with the following components:

- Explorer:** Shows a project structure for 'CSA0904' with folders 'CO1-AT1', 'CO1-AT2', 'CO4-AT1', 'CO4-AT2', and 'CO5-ClassActivity'. The 'CO5-ClassActivity' folder is expanded, showing files like 'CopyFileContent.java', 'CountFileStats.java', 'CreateAndWriteFile...', 'destination.txt', 'output.txt', 'ReadAndDisplayFile...', 'RecreateFile.java', 'sample_file.txt', and 'source.txt'.
- Editor:** Displays the source code for 'ReadAndDisplayFile.java'. The code imports `java.io.BufferedReader`, `java.io.File`, `java.io.FileReader`, and `java.io.IOException`. It defines a `public class ReadAndDisplayFile` with a `main` method that checks if the file 'output.txt' exists. If it does, it reads the content line by line using a `BufferedReader` and prints it. If it doesn't exist, it prints an error message.
- Terminal:** Shows the execution output of the command `java ReadAndDisplayFile.java`. The output is as follows:

```
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity> java ReadAndDisplayFile.java
--- Reading Content from: output.txt ---
1: Hello, World!
2: This file was created and written using Java File I/O.
3: Writing data completed successfully.
--- End of file ---
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO5-ClassActivity>
```